Milestone 3 Technical Memo

Household appliance energy consumption: baseline comparison

To: CSCI 3052U: Machine Learning I

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1. Purpose and current status

This milestone implements the initial model comparison proposed in Milestone 2 [2]. We estimate appliance energy use in Wh for the same 10 minute period as the inputs. This is regression using available measurements, not a forecast of future consumption. The median benchmark currently has the lowest validation MAE. None of the trained models improves on both benchmarks by our main metric. Tuning and feature group comparisons remain planned work.

2. Data and evaluation method

The UCI dataset contains 19,735 observations from one house in Belgium, collected from January to May 2016 [1]. The file has no missing values, duplicate timestamps, or gaps in its 10 minute spacing. We exclude the target and both random control columns, replace the timestamp with hour and day of week, and retain 27 predictors. Lighting is included on the assumption that its measurement is available for the same period.

The chronological split contains 13,814 training rows, 2,960 validation rows, and 2,961 test rows. Five expanding window folds use only the training block. Every fold fits earlier rows and evaluates the next 2,302 rows. StandardScaler is fitted inside the Linear Regression and kNN pipelines, using only the training rows for that fit. The test block has not been used for model evaluation.

Starting settings are kNN with 5 neighbours; a decision tree with minimum leaf size 5; and a random forest with 100 trees and minimum leaf size 5. Both tree models have no maximum depth and use seed 42. Fresh pipeline copies are fitted on the full training block for validation. The benchmarks predict the training mean, 98.775 Wh, or median, 60 Wh, for every validation row. No parameter search has been run.

3. Overall validation results

Model	MAE (Wh)	RMSE (Wh)	R^2
Median benchmark	42.605	98.108	-0.131
kNN	53.035	102.139	-0.226
Mean benchmark	53.968	92.394	-0.003
Linear Regression	55.235	87.077	0.109
Random Forest	61.131	95.912	-0.081
Decision Tree	72.584	127.180	-0.901

Lower MAE and RMSE are better. MAE is the primary metric; RMSE gives more weight to large errors. The median benchmark wins on MAE, while Linear Regression has the lowest RMSE and is the only method with positive R^2. A low constant prediction works well for many ordinary periods but misses large peaks. These are initial validation results, not final test results.

4. Fold results and variability

Mean fold MAE was 65.225 Wh for Linear Regression, 71.933 Wh for kNN, 73.529 Wh for Random Forest, and 86.582 Wh for the decision tree. The corresponding sample standard deviations were 12.096, 6.986, 14.526, and 14.830 Wh. Linear Regression ranked first across folds, but kNN won among trained models on validation MAE. Different periods and training sizes can change rankings. Overlapping training rows mean these standard deviations are descriptive, not confidence intervals.

5. Errors during high consumption

We define high consumption using the training target's 95th percentile, 340 Wh. This identifies 102 validation periods, or 3.4% of that block. The proportion can differ because the cutoff comes from training data. High values do not necessarily mean sudden jumps.

Model	High use MAE (Wh)	Underestimated (%)
Decision Tree	301.124	96.1
Random Forest	302.896	100.0
Linear Regression	326.721	100.0
kNN	374.667	99.0
Mean benchmark	384.656	100.0
Median benchmark	423.431	100.0

The decision tree has the lowest subset MAE despite its worst overall MAE. Random Forest has the lowest subset RMSE, 335.692 Wh. All methods usually underestimate consumption. On April 30 at 10:00, actual consumption was 840 Wh; Random Forest predicted 86.5 Wh. All subset R^2 scores are negative: squared errors exceed those of predicting the subset's actual average. That reference average is unavailable in advance.

6. Interpretation, limits, and next steps

Greater model complexity has not produced better overall MAE with these settings. The current results do not establish our final success criterion, which requires test MAE below both benchmarks. One house and part of one year also limit any claim about other homes or seasons. The supplied weather data includes interpolated hourly readings [1]; live use would need features built only from information available at prediction time.

Next, tune kNN and tree settings within the five training folds, compare environmental and time feature groups, and repeat the comparison without lighting. Inspect large errors without removing genuine peaks to improve scores. Since validation results have already been examined, avoid repeatedly changing settings just to improve that block. Freeze the final model before evaluating the test block once.

7. Reproducibility

The notebook [3] records the dataset hash, package versions, settings, and timing. Seven CSV exports retain fold, validation, and high consumption predictions and metrics; a JSON file records run details.

References

[1] Candanedo, L. (2017). Appliances Energy Prediction. UCI Machine Learning Repository. CC BY 4.0. doi.org/10.24432/C5VC8G

[2] Project team. (2026). Milestone 2: Dataset and Problem Approval. CSCI 3052U.

[3] Project team. (2026). MachineLearningProject.ipynb. Validated run, October 18, 2026.